

# Dimension-Controlled Linear Probing of Fashion-MNIST Convolutional Networks

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**Abstract**—Classifier accuracy does not reveal which distinctions are linearly accessible in a network’s hidden representations. Using Fashion-MNIST and frozen linear probes, I test whether footwear and tops, groupings absent from ten-class CNN training, can be recovered with few labels. Pixels, random CNN features, and trained activations are compared after matched Gaussian projection to 32 dimensions, across budgets of 10–200 images and 50 matched runs. With only 10 labels, trained FC features reached macro-F1 scores of 0.9803 for footwear and 0.9263 for tops, outperforming both baselines. Random features were nonetheless informative, especially for footwear. A second experiment compares ten-class supervision with footwear/non-footwear and tops/non-tops objectives. Coarse training made aligned boundaries highly accessible but did not consistently improve internal distinctions over random features, performing worse for footwear and only slightly better for tops; fine-label features remained strongest. Thus, in this setting, fine-label training exposed broader groupings not directly supervised, whereas coarse objectives offered no consistent benefit for their internal structure.

**Index Terms**—linear probes, representation learning, sample efficiency, Fashion-MNIST, convolutional neural networks

## I. Introduction

Convolutional neural networks are normally evaluated through the accuracy of their final classifier. This leaves a more diagnostic question unanswered: what information is already available to a simple downstream learner at each hidden layer? Linear probes attempt this question by fitting an independent linear classifier to frozen activations, without changing the network itself [2]. A high probe score indicates that a target concept is linearly accessible, although it does not prove that the CNN’s own classifier uses that concept.

Fashion-MNIST contains 70,000 grayscale 28×28 images in ten clothing classes, with an official split of 60,000 training and 10,000 test images [1]. Though not natively provisioned, its low level labels quite conveniently support grouping at a more general semantic level: for example, sandal, sneaker, and ankle boot form a footwear superclass, while T-shirt/top, pullover, coat, and shirt form a tops superclass. This makes it a useful testbed for studying how supervised objectives can shape representations. Fashion-MNIST therefore provides a compact setting in which to study the decodability of broader semantic categories. We use raw pixels as a baseline for how accessible these categories are in the input itself. Be-

cause even randomly initialized CNNs can produce useful features [3], we additionally compare random and trained CNN representations to distinguish the contribution of the architecture from that of supervised learning.

This project asks two questions. First, can superclass distinctions absent from CNN training be decoded from its hidden layers using only a few probe labels, and where are they most accessible? Second, does explicit superclass training also improve distinctions within that superclass, or does it primarily strengthen the trained boundary? Experiment 1 probes footwear and tops throughout a ten-class CNN, using pixels and a random CNN as baselines. Experiment 2 compares ten-class supervision with footwear/non-footwear and tops/non-tops supervision. All comparisons use matched labeled examples and a common 32-dimensional feature budget.

## II. Methods

### A. Data, Architecture, and Training

Pixels were scaled to  $[0,1]$  without augmentation. A stratified split assigned 50,000 official training images to CNN fitting and 10,000 to validation. The official 10,000-image test set was held out until all models and probes had been selected. The CNN contains two  $3 \times 3$  convolutional layers, producing 16 and 32 feature maps, respectively. Each uses ReLU and  $2 \times 2$  max pooling, followed by a 128-unit ReLU layer and a classifier. Flattened activation dimensions are 3,136 (Conv1), 1,568 (Conv2), and 128 (FC).

Models were trained for 10 epochs with cross-entropy loss, Adam, learning rate  $10^{-3}$ , and batch size 64. The checkpoint with the lowest validation loss was retained. All reported probes used one seed-42 realization of each CNN condition. Each probe condition was evaluated using ten labeled-data samples with and five Gaussian random projections, yielding 50 matched runs. Reported means and standard deviations therefore reflect variation in probe sampling and projection, not variation across CNN training runs. The main model was trained on the original ten Fashion-MNIST classes. In the extension, the random, fine-label, and coarse-label networks began with identical shared the same initial feature-extractor weights. The fine-label network was trained on the ten origi-

nal classes, whereas the coarse-label network predicted footwear (classes 5, 7, and 9) versus non-footwear.

## B. Dimension-Controlled Linear Probes

I compared raw pixels and frozen Conv1, Conv2, and FC activations from random and trained CNNs. Every representation was mapped to 32 dimensions by a fixed Gaussian random projection. Random and trained representations from the same layer received identical projection matrices. Five projection seeds (0-4) were crossed with ten nested sampling repeats (sampling seeds 42-51).

L2-regularized logistic regressions were trained with exact total label budgets  $n \in \{10, 20, 30, 50, 80, 110, 150, 200\}$ . Extracted representations were standardized feature-wise using statistics computed exclusively on the probe-training split and then applied unchanged to the validation and test splits. The inverse regularization strength  $C \in \{0.01, 0.1, 1\}$  was selected by macro-F1 on a fixed 2,000-image validation subset. All probe models were fitted and serialized before test-set evaluation. For each task and representation condition, results report the mean and sample standard deviation over the 50 matched sampling/projection runs.

The main tasks in Experiment 1 were binary footwear and tops recognition. Samples were nested and stratified over all ten original classes. Experiment 2 evaluated footwear and tops separately, using seven and eleven task-specific probes, respectively. For each superclass, these comprised one multiclass task, all pairwise contrasts, and one class-versus-non-superclass task per class. Training examples were distributed as evenly as possible across the target classes. Macro-F1 was the primary metric; accuracy, balanced accuracy, per-class F1, and confusion matrices were also stored. Split indices, checkpoint hashes, environment metadata, raw results, and fitted probe bundles are retained for reproducibility; code and artifacts will be published at [PUBLIC-GITLAB-URL].

## III. Results and Discussion

### A. Experiment 1: Layer-Wise Sample Efficiency

The ten-class CNN showed mild overfitting. Training accuracy increased from 0.9155 at the selected epoch 6 to 0.9361 at epoch 10, while validation loss was lowest at epoch 6 (0.2499) and did not improve thereafter. Checkpoint selection therefore avoided using the later, more overfit model.

Fig. 1 shows that learned representations were most beneficial when labels were scarce. With only 10 probe-training images, trained FC features achieved  $0.9803 \pm 0.0192$  macro-F1 for footwear, compared with  $0.9168 \pm 0.0521$  for random FC features and  $0.8754 \pm 0.0627$  for pixels. For tops, the corresponding scores were  $0.9263 \pm 0.0223$ ,  $0.7808 \pm 0.0595$ , and  $0.7837 \pm 0.0424$ . At 200 labels, trained FC features remained strongest, reaching 0.9974 for footwear and 0.9625 for tops.

The strong random-CNN baseline suggests that the convolutional architecture contributes to the gain over pixels. Even without training, convolution encodes locality, while ReLU and pooling produce nonlinear features with some tolerance to small translations. Mapping every representation to 32 dimensions prevents feature count alone from explaining the differences. Nevertheless, trained FC features performed best across all label budgets, with the largest advantages at low  $n$  and for tops. This pattern is consistent with supervised learning making task-relevant distinctions more linearly accessible.

### B. Experiment 2: Training Objective Shapes Transfer

Experiment 2 asked whether learning a coarse superclass boundary also improves distinctions within that superclass. It did not do so consistently. At 200 labels, coarse FC was below random FC on three-way footwear (0.8505 versus 0.8635) and averaged slightly lower across the footwear pairwise contrasts (0.9200 versus 0.9278). For tops, however, coarse FC exceeded random FC on the four-way task (0.6185 versus 0.5771) and across the pairwise contrasts (0.8375 versus 0.8174). This difference may reflect that tops-versus-non-tops was the harder coarse task (0.9809 versus 0.9992 test accuracy), requiring richer features that also preserved more within-superclass structure. Fine-label FC remained substantially stronger on both multiway tasks, reaching 0.9585 for footwear and 0.7973 for tops.

As expected, coarse FC performed strongly on the aligned class-versus-non-superclass probes. With only 10 labels, it averaged 0.9942 for footwear and 0.9392 for tops, compared with 0.9401 and 0.8799 for fine FC. This boundary advantage is unsurprising; the more informative result is that transfer within the superclass was category-dependent and could fall below the untrained baseline. Fig. 2 summarizes both effects.

### C. Limitations

The analysis uses one seed-42 training run per CNN condition; the 50 probe runs therefore capture sampling and projection variability, not variation across learned backbones. Fashion-MNIST is small and visually regular, and near-ceiling performance on several superclass-boundary tasks may obscure differences between representations. Linear probes measure only linear accessibility, while random projection may distort representations differently despite equalizing their dimensionality. Replication across training seeds, architectures, and a more challenging natural-image dataset would provide a stronger generalization test.

## IV. Conclusion

Experiment 1 showed that footwear and tops, although absent from the ten-class objective, were decodable with only 10 probe labels and most accessible at FC. Random features were informative, but trained FC features

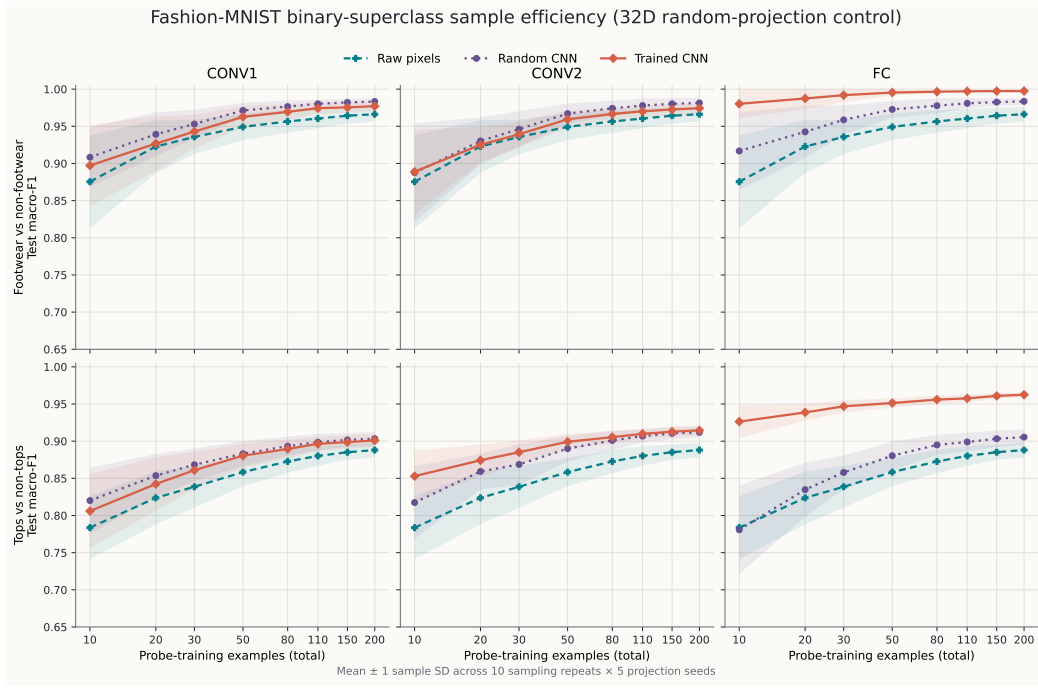


Fig. 1. Test macro-F1 as a function of the exact total probe-training budget. Lines show means and shaded regions show one sample standard deviation across 10 nested sampling repeats and five matched 32D Gaussian projections. The random CNN is a strong control, but supervised features, especially FC features, are more sample-efficient.

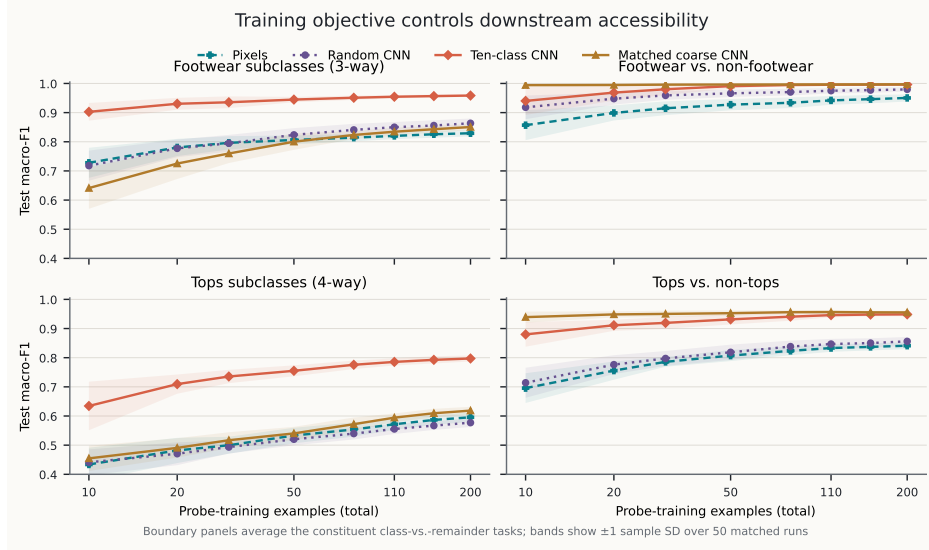


Fig. 2. Experiment 2, using 32D FC features. Rows show footwear and tops; columns show within-superclass decoding and the corresponding superclass boundary. Lines and shaded regions show mean  $\pm$  one sample standard deviation over the same 50 matched runs as Experiment 1.

performed best, especially for tops. Experiment 2 showed that coarse training learned its target boundary but did not consistently improve internal distinctions over random features, performing worse for footwear and only slightly better for tops.

## References

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